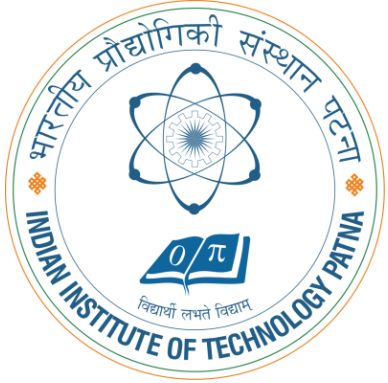


CS5102: FOUNDATIONS OF COMPUTER SYSTEMS

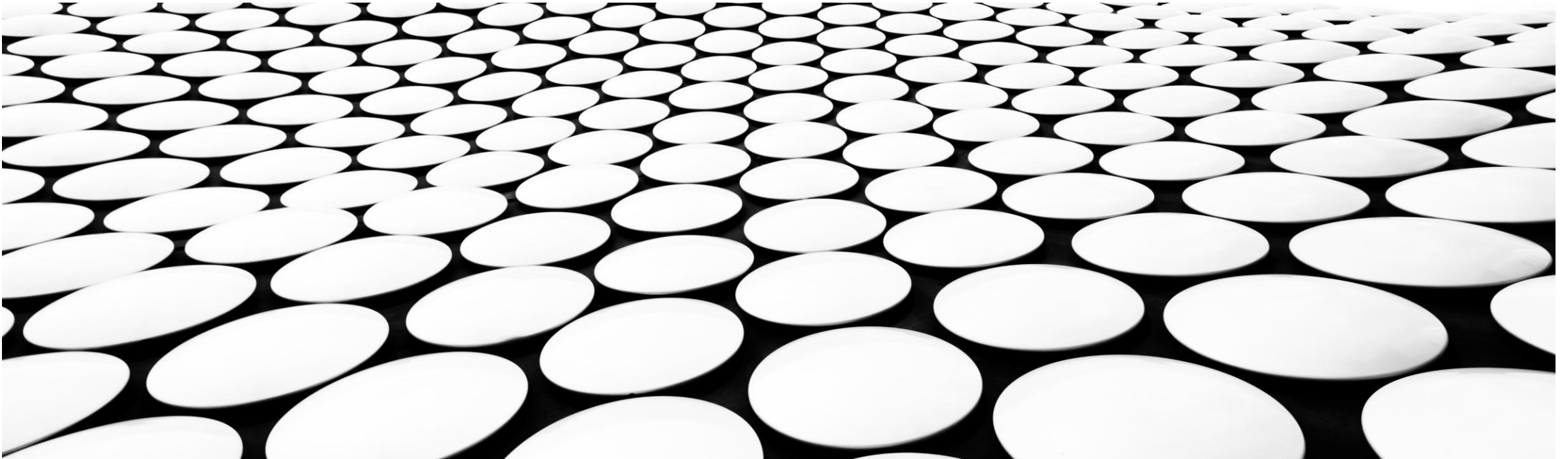


TOPIC-18 THREAD AND DISCUSSION ON OS

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RECAP

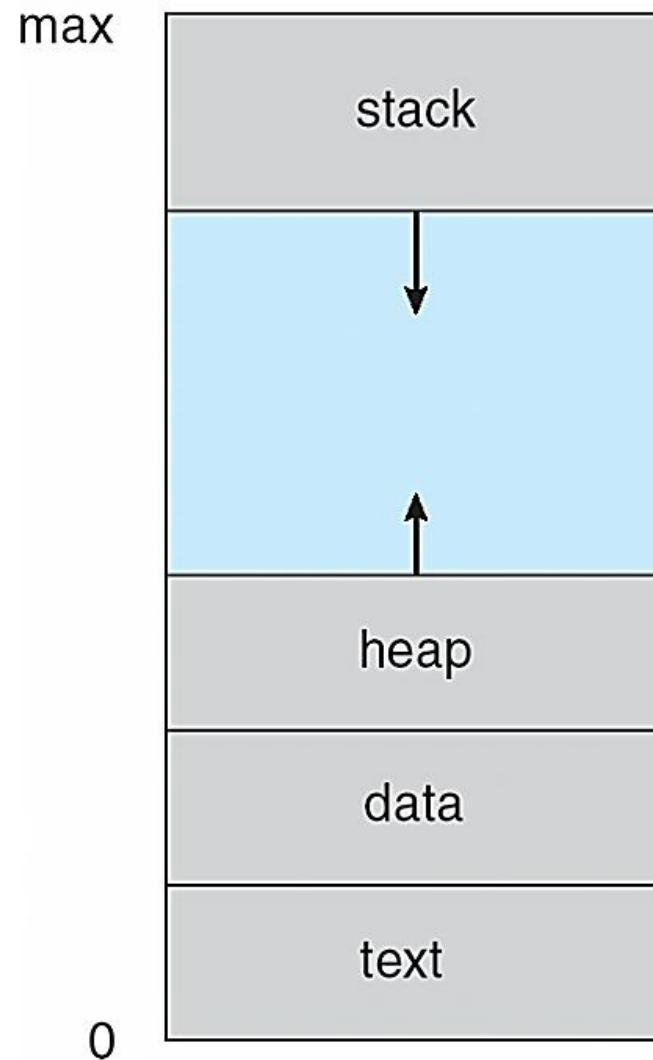
PROCESS CONCEPT

- An operating system executes a variety of programs:
 - Batch system – **jobs**
 - Time-shared systems – **user programs** or **tasks**
- Textbook uses the terms **job** and **process** almost interchangeably
- **Process** – a program in execution; process execution must progress in sequential fashion
- Multiple parts
 - The program code, also called **text section**
 - Current activity including **program counter**, processor registers
 - **Stack** containing temporary data
 - Such as *Function parameters, return addresses, local variables*
 - **Data section** containing global variables
 - **Heap** containing memory dynamically allocated during run time

PROCESS CONCEPT (CONTD.)

- Program is *passive* entity stored on disk (**executable file**), process is *active*
 - Program becomes process when executable file loaded into memory
- Execution of program started via GUI mouse clicks, command line entry of its name, etc
- One program can be several processes
 - Consider multiple users executing the same program

PROCESS IN MEMORY

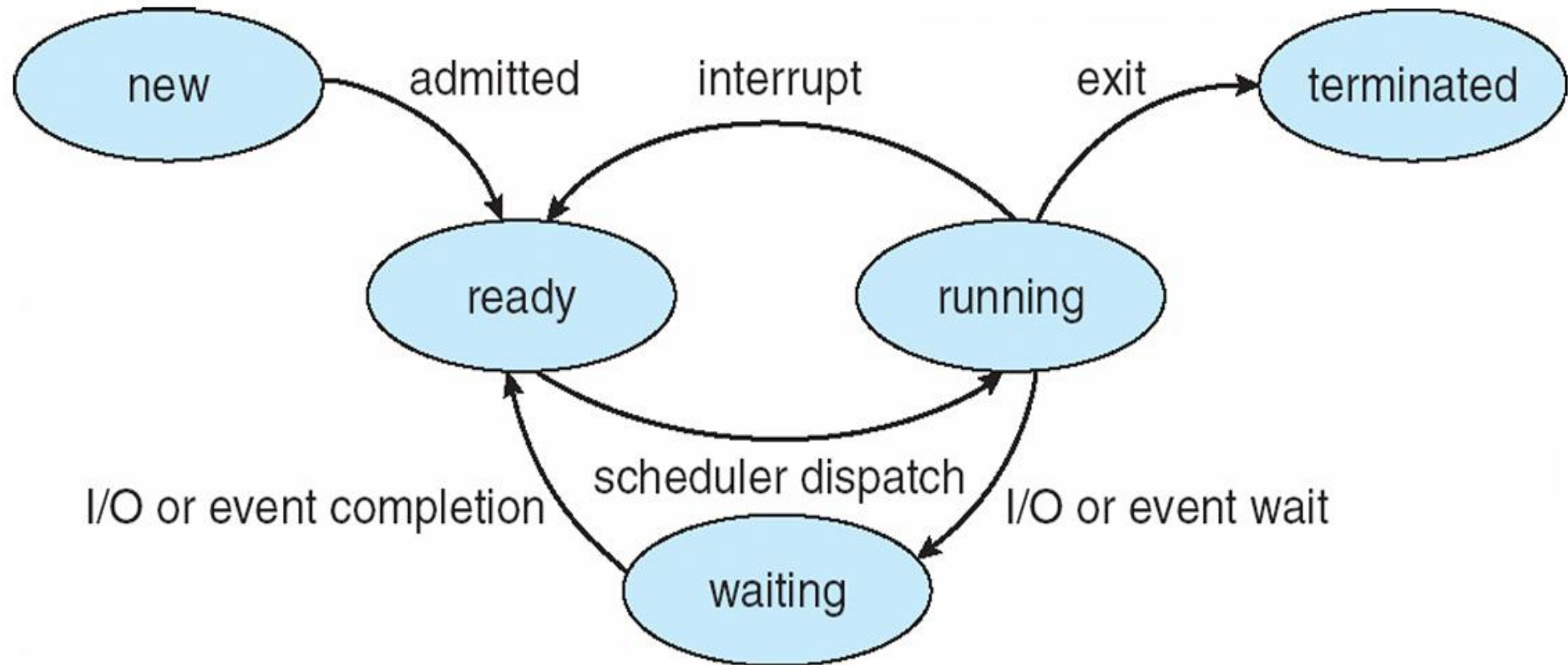


PROCESS STATE



- As a process executes, it changes **state**
 - **new**: The process is being created
 - **running**: Instructions are being executed
 - **waiting**: The process is waiting for some event to occur
 - **ready**: The process is waiting to be assigned to a processor
 - **terminated**: The process has finished execution

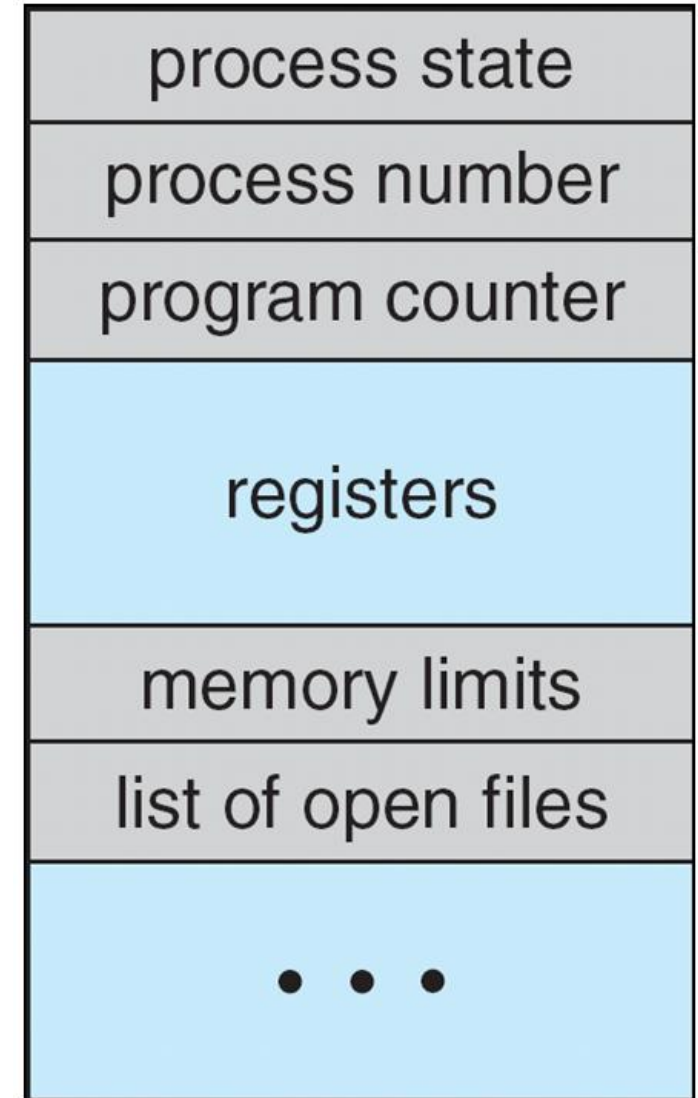
DIAGRAM OF PROCESS STATE



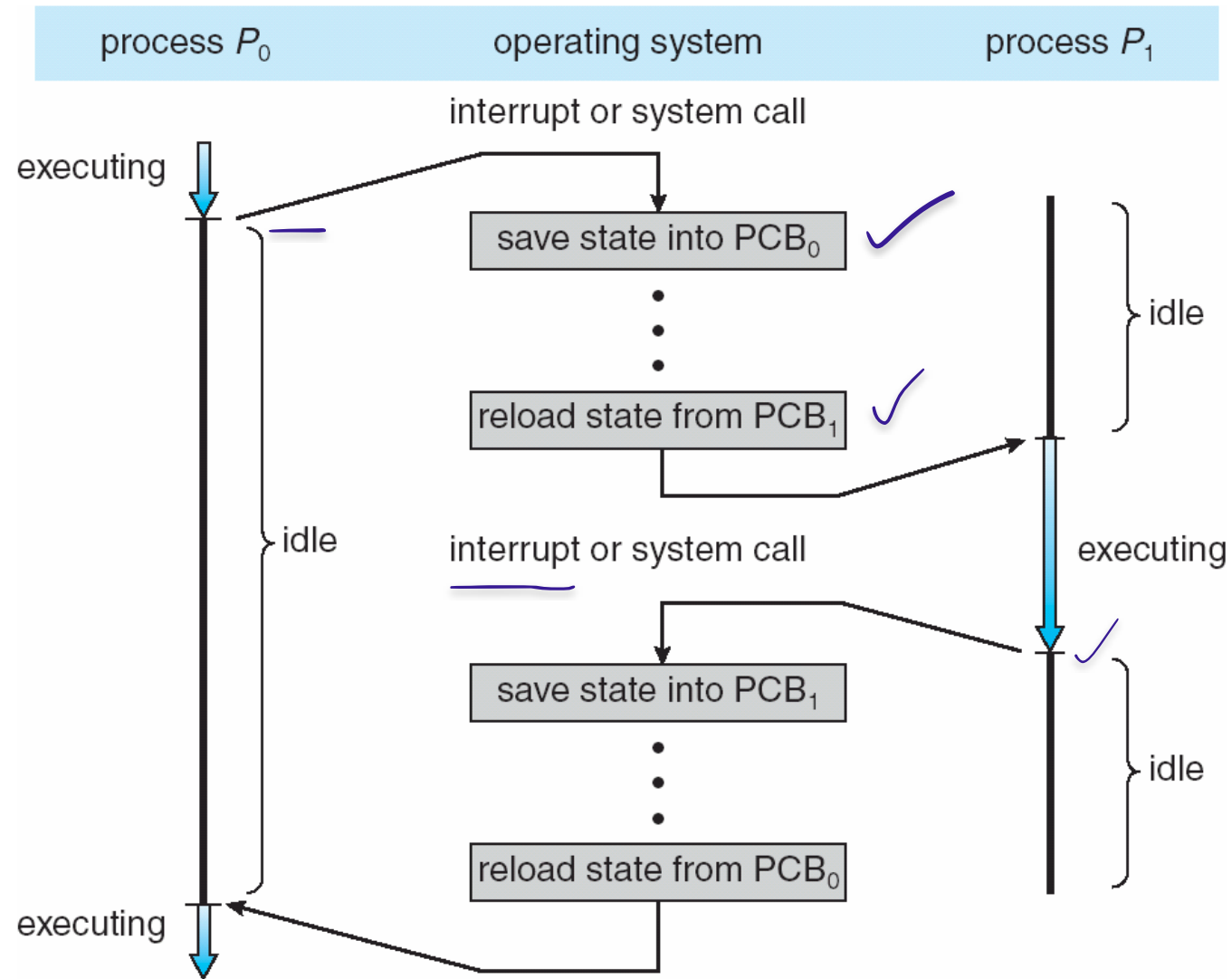
PROCESS CONTROL BLOCK (PCB)

Information associated with each process
(also called **task control block**)

- Process state – running, waiting, etc
- Program counter – location of instruction to next execute
- CPU registers – contents of all process-centric registers – *Accumulators (store intermediate logical or arithmetic data), indexed registers, stack pointer, general purpose registers*
- CPU scheduling information- priorities, scheduling queue pointers
- Memory-management information – memory allocated to the process
- Accounting information – CPU used, clock time elapsed since start, time limits
- I/O status information – I/O devices allocated to process, list of open files



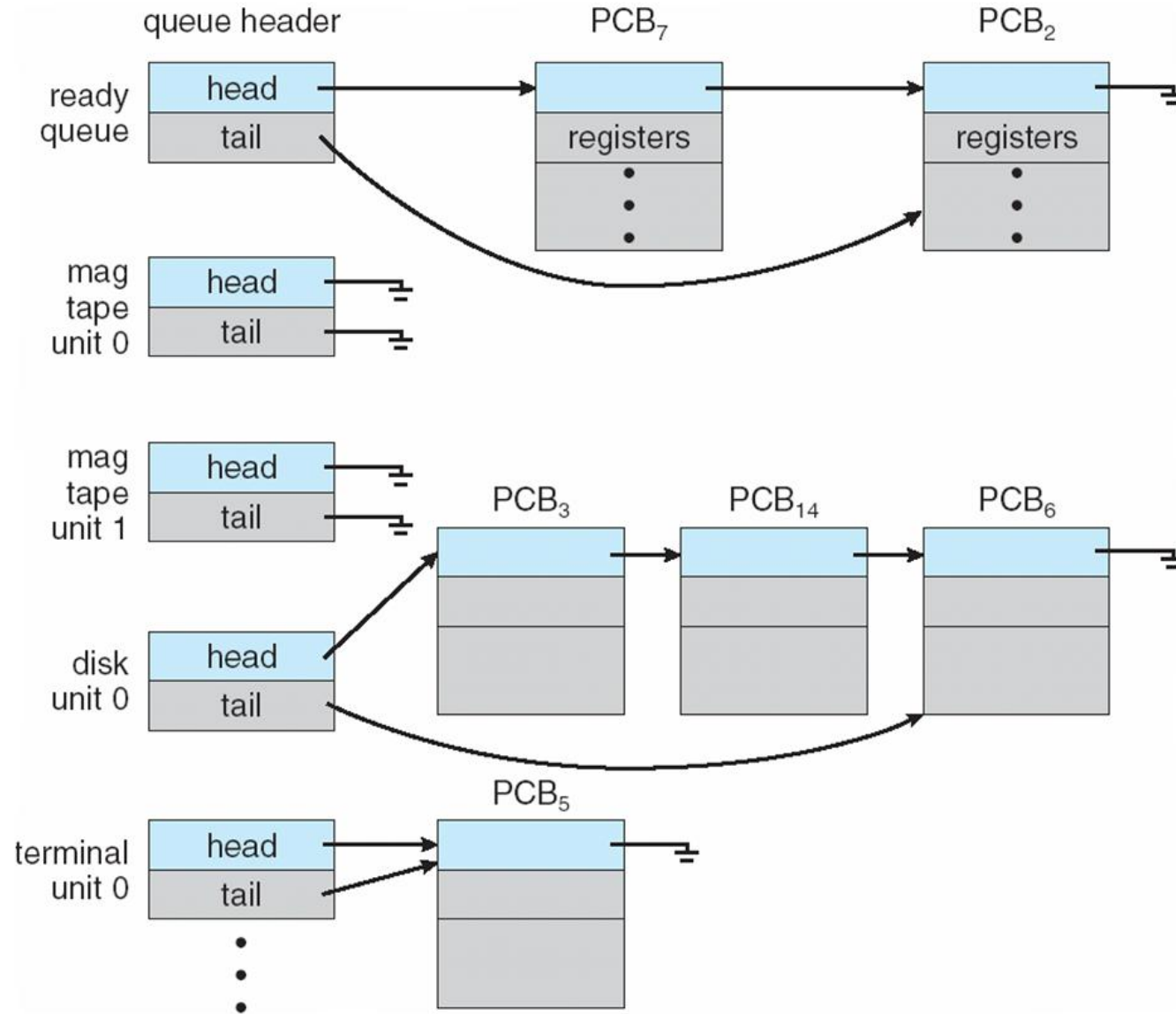
CPU SWITCH FROM PROCESS TO PROCESS



PROCESS SCHEDULING

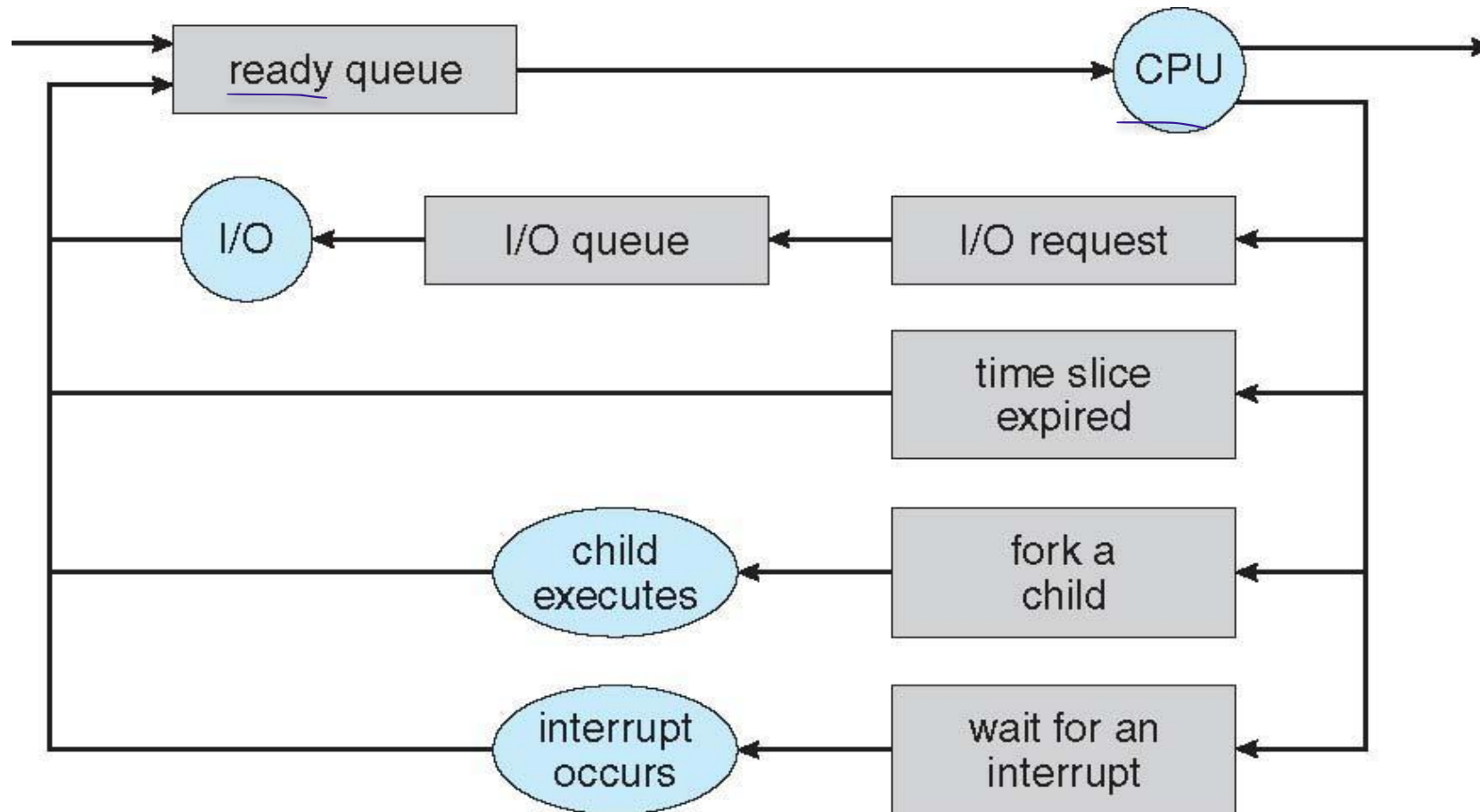
- Maximize CPU use, quickly switch processes onto CPU for time sharing
- **Process scheduler** selects among available processes for next execution on CPU
- Maintains **scheduling queues** of processes
 - **Job queue** – set of all processes in the system
 - **Ready queue** – set of all processes residing in main memory, ready and waiting to execute
 - **Device queues** – set of processes waiting for an I/O device
 - Processes migrate among the various queues

READY QUEUE AND VARIOUS I/O DEVICE QUEUES



REPRESENTATION OF PROCESS SCHEDULING

- Queueing diagram represents queues, resources, flows



SCHEDULERS

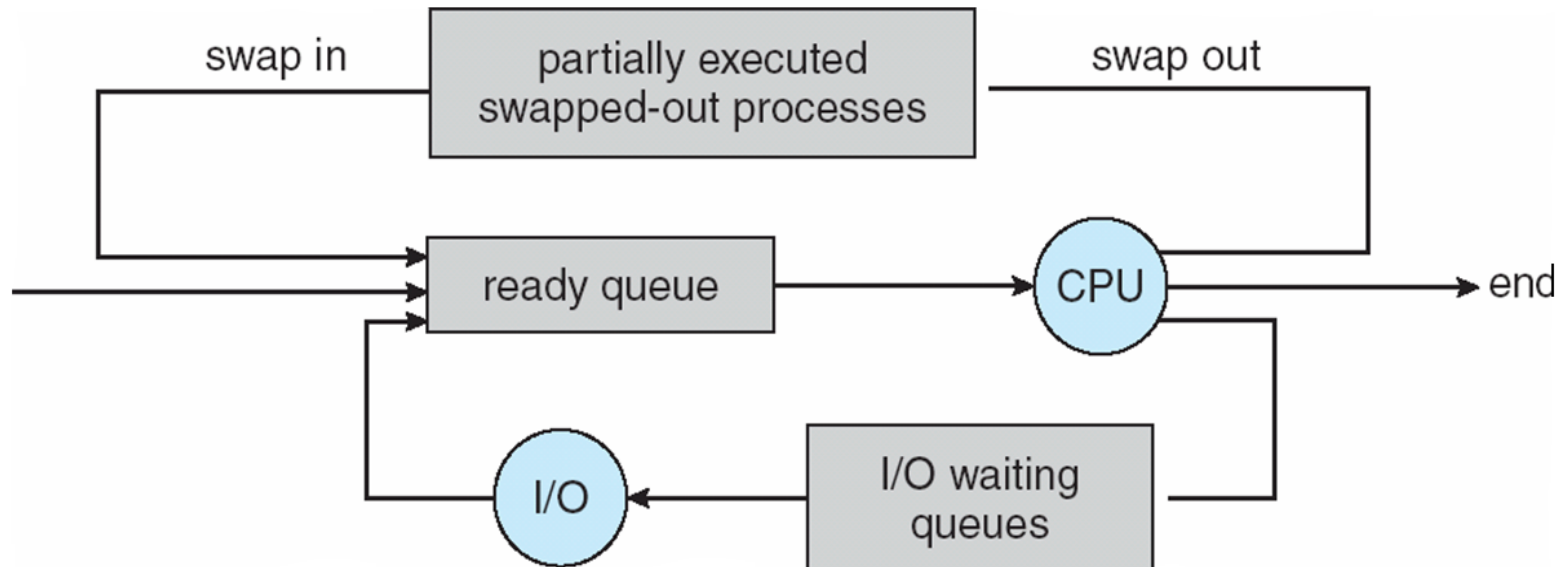
- **Short-term scheduler** (or **CPU scheduler**) – selects which process should be executed next and allocates CPU
 - Sometimes the only scheduler in a system
 - Short-term scheduler is invoked frequently (milliseconds) $\Rightarrow$ (must be fast)
- **Long-term scheduler** (or **job scheduler**) – selects which processes should be brought into the ready queue
 - Long-term scheduler is invoked infrequently (seconds, minutes) $\Rightarrow$ (may be slow)
 - The long-term scheduler controls the **degree of multiprogramming**
- Processes can be described as either:
 - **I/O-bound process** – spends more time doing I/O than computations, many short CPU bursts
 - **CPU-bound process** – spends more time doing computations; few very long CPU bursts
- Long-term scheduler strives for good *process mix*

The number of processes in the memory

If the degree of multiprogramming is stable, then the average rate of process creation must be equal to the average departure rate of process leaving the system

ADDITION OF MEDIUM TERM SCHEDULING

- **Medium-term scheduler** can be added if degree of multiple programming needs to decrease
 - Remove process from memory, store on disk, bring back in from disk to continue execution:
swapping



Some systems may have no long-term scheduler or a minimal use of it

e.g., UNIX and Microsoft Windows often have no long-term scheduler

Every process put in to the memory for the short-term scheduler

CONTEXT SWITCH

- When CPU switches to another process, the system must **save the state** of the old process and load the **saved state** for the new process via a **context switch**
- **Context** of a process represented in the PCB
- Context-switch time is overhead; the system does no useful work while switching
 - The more complex the OS >> the longer the context switch
- Time dependent on hardware support
 - Some hardware provides multiple sets of registers per CPU >> multiple contexts loaded at once

OPERATIONS ON PROCESSES

- System must provide mechanisms for:
 - process creation,
 - process termination,
 - and so on as detailed next

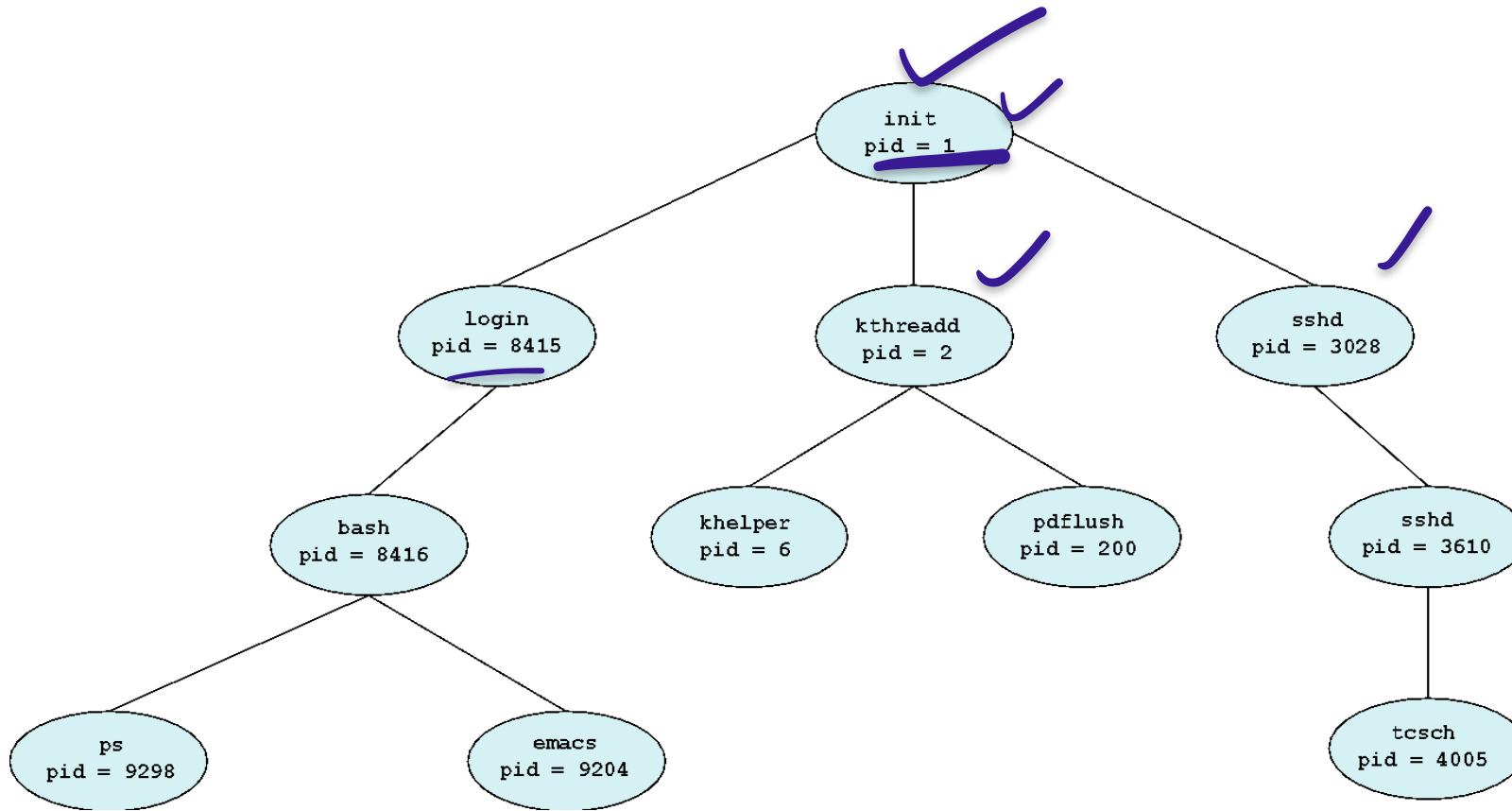
PROCESS CREATION

- **Parent** process create **children** processes, which, in turn create other processes, forming a **tree** of processes
- Generally, process identified and managed via a **process identifier (pid)**
- Resource sharing options
 - Parent and children share all resources
 - Children share subset of parent's resources
 - Parent and child share no resources
- Execution options
 - Parent and children execute concurrently
 - Parent waits until children terminate

e.g., CPU time, memory, files, I/O devices

Restricting a child process to a subset of parent's resources prevent any process from overloading the system by creating too many child processes

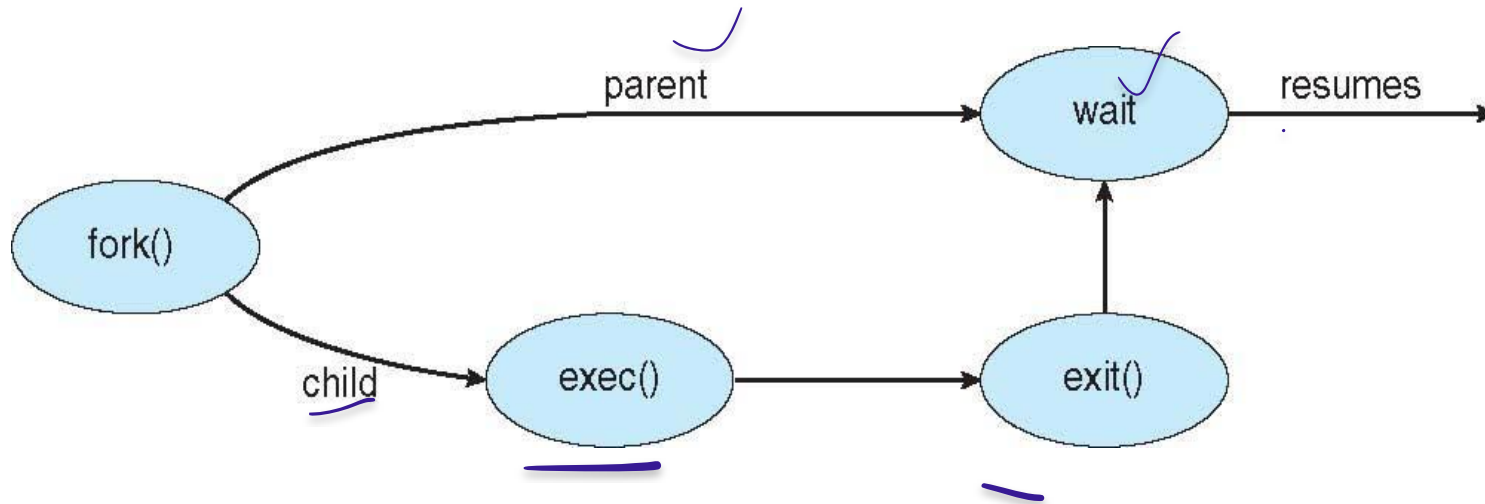
A TREE OF PROCESSES IN LINUX



- Root parent process for all user processes
- Once the system is booted, the *init* process can also create various user processes such as web or print server, ssh server, etc.
- Two child processes – *kthreadd* and *sshd*
- *kthreadd* is responsible for creating additional processes that perform task on behalf of the kernel.
- *sshd* process is responsible for managing the clients that connect to the system by using ssh (secure shell)

PROCESS CREATION (CONT.)

- Address space
 - Child duplicate of parent
 - Child has a program loaded into it
- UNIX examples
 - **fork()** system call creates new process
 - **exec()** system call used after a fork() to replace the process' memory space with a new program



C PROGRAM FORKING SEPARATE PROCESS

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
        fprintf(stderr, "Fork Failed");
        return 1;
    }
    else if (pid == 0) { /* child process */
        execlp("/bin/ls", "ls", NULL);
    }
    else { /* parent process */
        /* parent will wait for the child to complete */
        wait(NULL);
        printf("Child Complete");
    }

    return 0;
}
```

PROCESS TERMINATION

- Process executes last statement and then asks the operating system to delete it using the **exit()** system call.
 - Returns status data from child to parent (via **wait()**)
 - Process' resources are deallocated by operating system
- Parent may terminate the execution of children processes using the **abort()** system call. Some reasons for doing so:
 - Child has exceeded allocated resources
 - Task assigned to child is no longer required
 - The parent is exiting and the operating systems does not allow a child to continue if its parent terminates

PROCESS TERMINATION

- Some operating systems do not allow child to exist if its parent has terminated. If a process terminates, then all its children must also be terminated.
 - ✓ **cascading termination**
 - All children, grandchildren, etc. are terminated.
 - The termination is initiated by the operating system.
- The parent process may wait for termination of a child process by using the **wait()** system call. The call returns status information and the pid of the terminated process
pid = wait(&status);
- Terminated, but whose parent did not invoke **wait()** process is a **zombie**
- If parent terminated without invoking **wait**, process is an **orphan**

THREADS

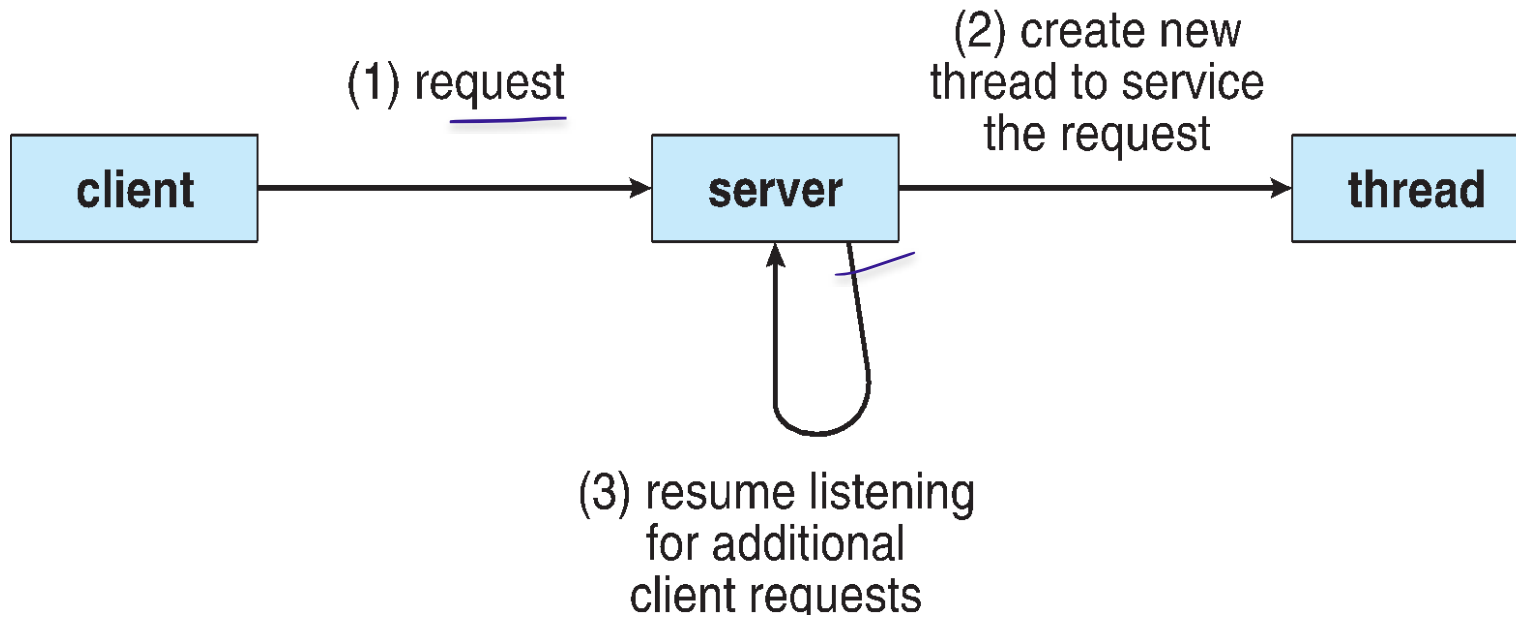
THREADS

- So far, process has a single thread of execution
 - Consider having multiple program counters per process
 - Multiple locations can execute at once
- Multiple threads of control >> **threads**
- Must then have storage for thread details, multiple program counters in PCB

MOTIVATION

- Most modern applications are multithreaded
- Threads run within application
- Multiple tasks with the application can be implemented by separate threads
 - Update display
 - Fetch data
 - Spell checking
 - Answer a network request
- Process creation is heavy-weight while thread creation is light-weight
- Can simplify code, increase efficiency
- Kernels are generally multithreaded

MULTITHREADED SERVER ARCHITECTURE



BENEFITS

- **Responsiveness** – may allow continued execution if part of process is blocked, especially important for user interfaces
- **Resource Sharing** – threads share resources of process, easier than shared memory or message passing
- **Economy** – cheaper than process creation, thread switching lower overhead than context switching
- **Scalability** – process can take advantage of multiprocessor architectures

MULTICORE PROGRAMMING

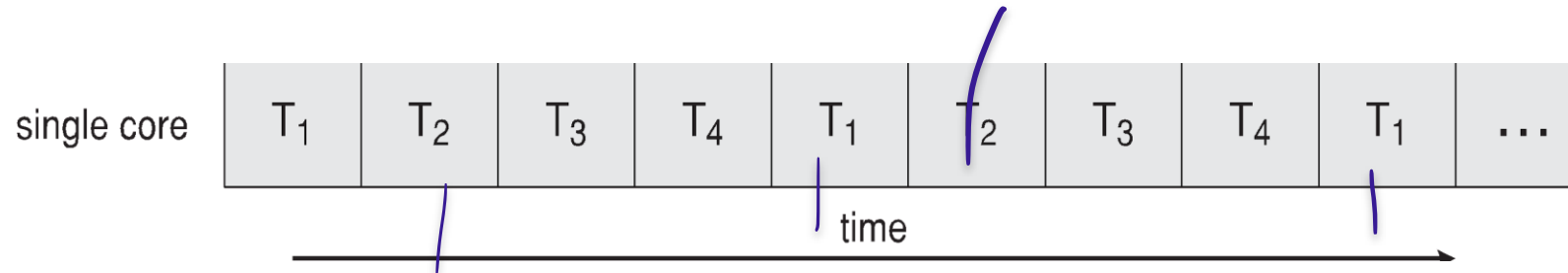
- **Multicore or multiprocessor** systems putting pressure on programmers, challenges include:
 - **Dividing activities**
 - **Balance**
 - **Data splitting**
 - **Data dependency**
 - **Testing and debugging**
- **Parallelism** implies a system can perform more than one task simultaneously ✓
- **Concurrency** supports more than one task making progress
 - Single processor / core, scheduler providing concurrency

MULTICORE PROGRAMMING (CONTD.)

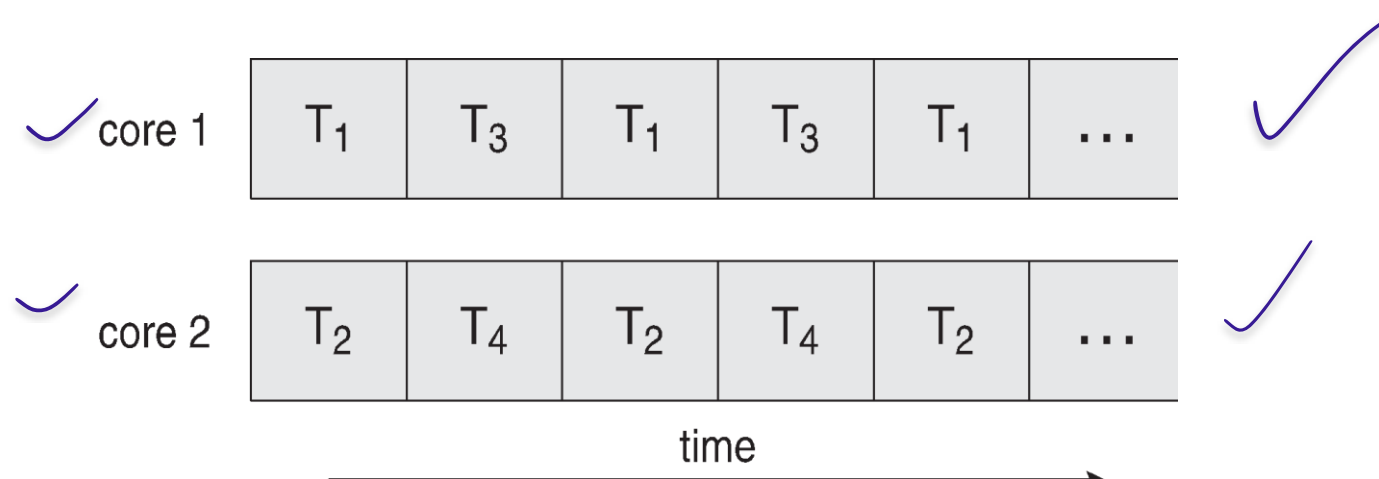
- Types of parallelism
 - **Data parallelism** – distributes subsets of the same data across multiple cores, same operation on each
 - **Task parallelism** – distributing threads across cores, each thread performing unique operation
- As the number of threads grows, so does architectural support for threading
 - CPUs have cores as well as *hardware threads*
 - Consider Oracle SPARC T4 with 8 cores, and 8 hardware threads per core

CONCURRENCY VS. PARALLELISM

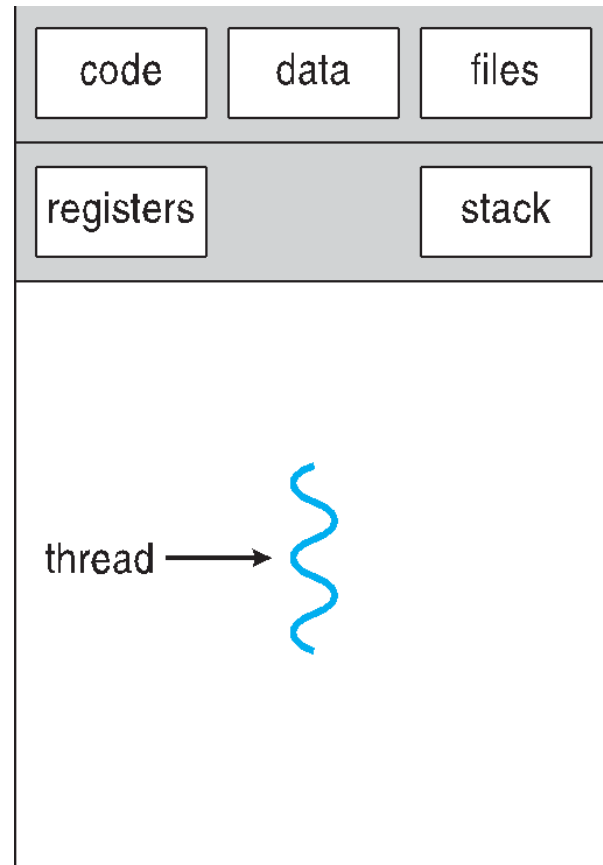
- Concurrent execution on single-core system:



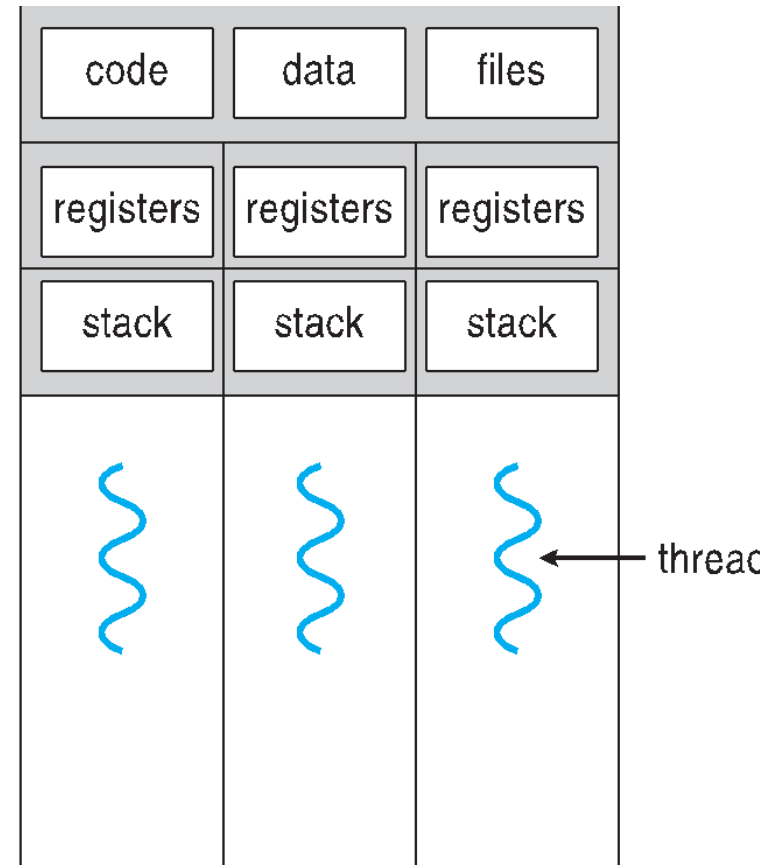
- Parallelism on a multi-core system:



SINGLE AND MULTITHREADED PROCESSES



single-threaded process



multithreaded process

USER THREADS AND KERNEL THREADS

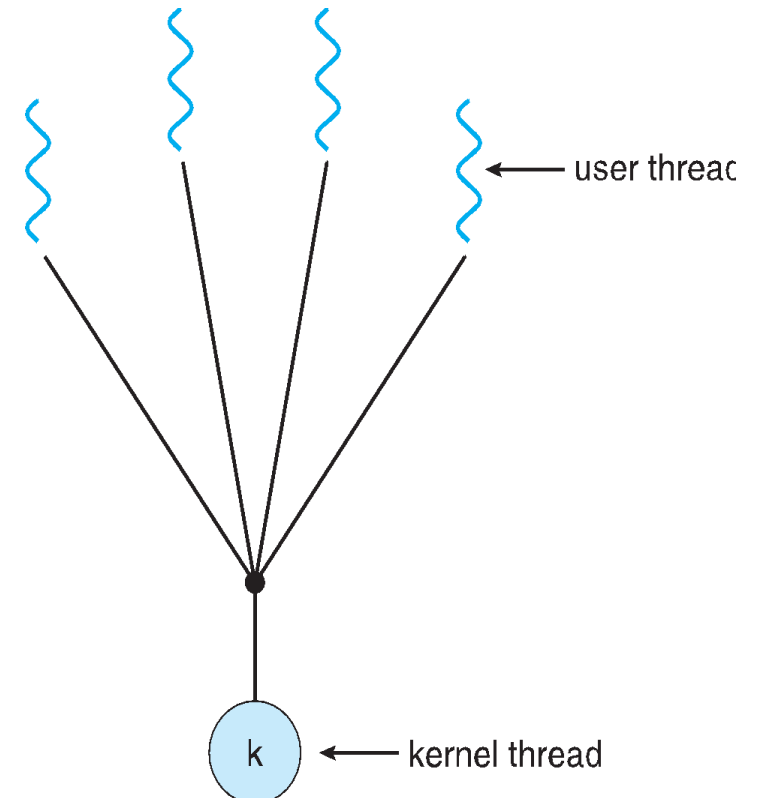
- **User threads** - management done by user-level threads library
- Three primary thread libraries:
 - POSIX **Pthreads**
 - Windows threads
 - Java threads
- **Kernel threads** - Supported by the Kernel
- Examples – virtually all general purpose operating systems, including:
 - Windows
 - Solaris
 - Linux
 - Tru64 UNIX
 - Mac OS X

MULTITHREADING MODELS

- Many-to-One
- One-to-One
- Many-to-Many

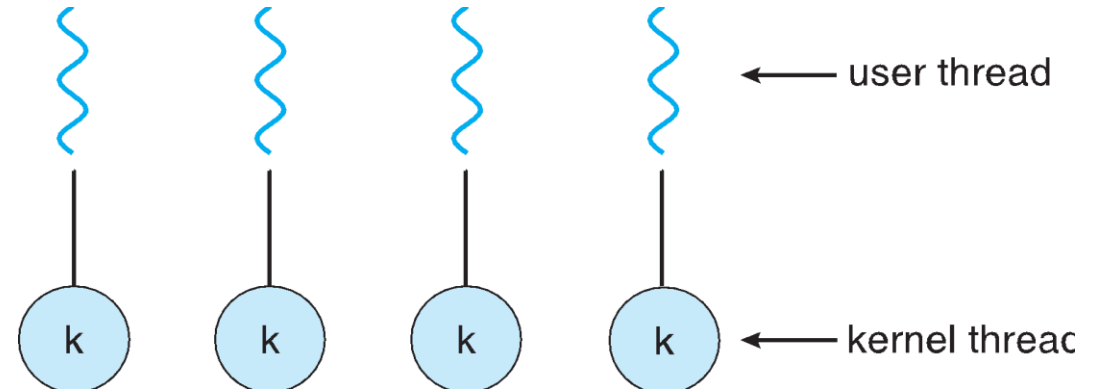
MANY-TO-ONE

- Many user-level threads mapped to single kernel thread
- One thread blocking causes all to block
- Multiple threads may not run in parallel on multicore system because only one may be in kernel at a time
- Few systems currently use this model
- Examples:
 - **Solaris Green Threads**
 - **GNU Portable Threads**



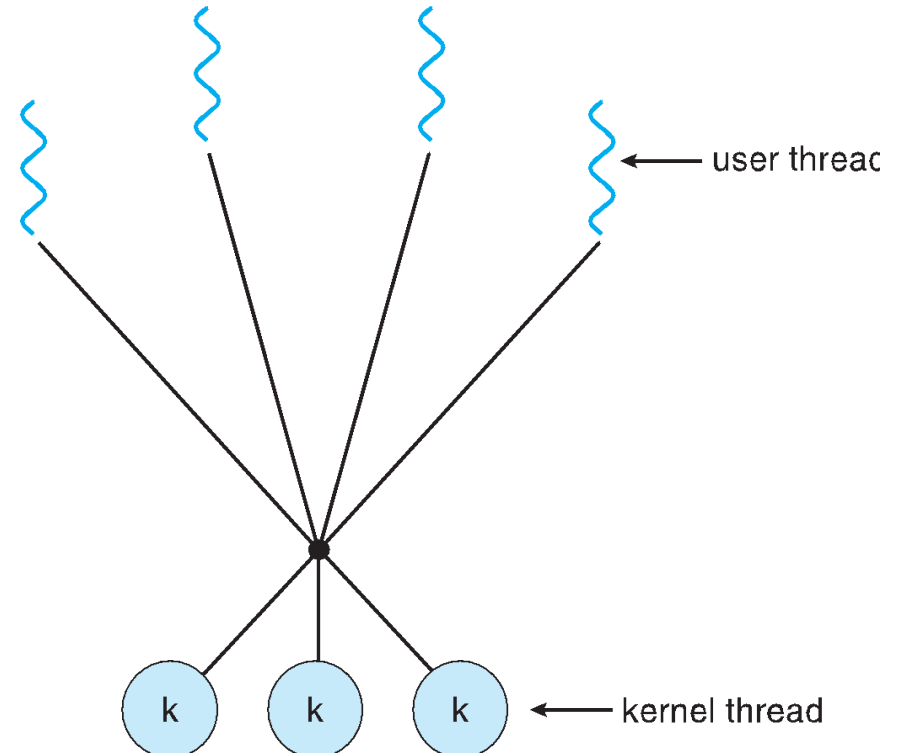
ONE-TO-ONE

- Each user-level thread maps to kernel thread
- Creating a user-level thread creates a kernel thread
- More concurrency than many-to-one
- Number of threads per process sometimes restricted due to overhead
- Examples
 - Windows
 - Linux
 - Solaris 9 and later



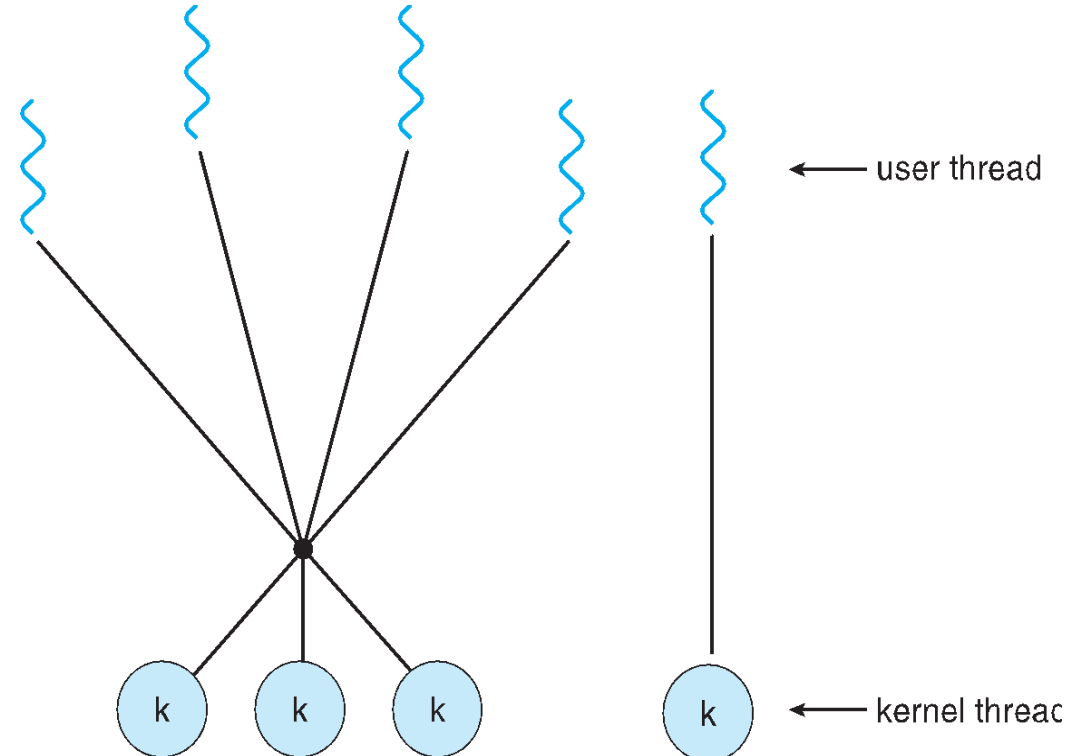
MANY-TO-MANY MODEL

- Allows many user level threads to be mapped to many kernel threads
- Allows the operating system to create a sufficient number of kernel threads
- Solaris prior to version 9
- Windows with the *ThreadFiber* package



TWO-LEVEL MODEL

- Similar to M:M, except that it allows a user thread to be **bound** to kernel thread
- Examples
 - IRIX
 - HP-UX
 - Tru64 UNIX
 - Solaris 8 and earlier



DISCUSSION

DISCUSSION ON OS

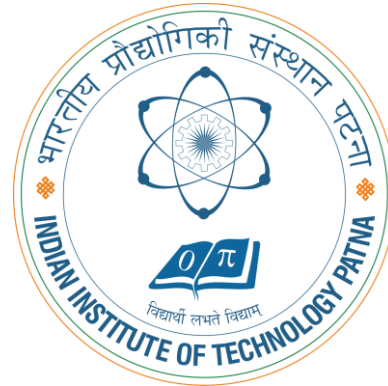
- What is an operating system? Give a simple definition.
- Name a few popular operating systems you have used.
- Why does a computer need an operating system?
- What happens when you start (boot) a computer?
- What are examples of input and output devices managed by an operating system?
- What is meant by "user interface" in an operating system?
- Name two examples of open-source operating systems.
- What is the difference between system software and application software?

DISCUSSION ON OS

- Can you name a device other than a computer that uses an operating system?
- What is the purpose of drivers in an operating system?
- What is multitasking in an operating system? Can you give a real-world example?
- What is a file system? Why is it important?
- How does an operating system manage memory (RAM)?
- Why is security important in an operating system? Give an example.

DISCUSSION ON OS

- What are the primary functions of an operating system (OS)?
- What is the difference between a kernel and a shell in an operating system?
- Compare and contrast a monolithic kernel with a microkernel architecture.
- Why is memory management crucial in an operating system?
- What role do system calls play in operating system architecture?
- How does an operating system ensure file system security and access control?
- What is the difference between a process and a thread?
- What are the advantages and challenges of using multi-threaded processes?



THANK YOU!